

Evidence Gap Analysis: Inability to Predict Monoculture Trends Due to Lack of Relevant Ecological Literature in Allowed Corpus

Input Question

Will we soon see the end of monocultures like wheat, maize, rice, and soy?

Domain

Ecology

Validation Status

needs_data

Problem Statement

The question asks for a prediction regarding the systemic collapse or replacement of major monoculture crops (wheat, maize, rice, soy) in the near future. The provided context highlights ecological downsides (soil degradation, biodiversity loss) but lacks empirical evidence on transition trends. Crucially, the allowed evidence catalog contains no relevant ecological or agricultural literature, making it impossible to formulate a data-driven hypothesis about the future of these systems based on the provided constraints.

Rationale

Strict adherence to the SAGE125 protocol requires that all factual claims be grounded in allowed EvidenceCards. The provided EvidenceCards (EV-Q113-922f1ca507ca9ae55ea64686, EV-Q113-9fbb7400332c8c3f75e4963c, EV-Q113-6eb7f4dea9363af7283d0969, EV-Q113-f8776fe39ba76542f19b29e8) pertain to Computer Vision, Astrophysics, Microbiology, and unrelated topics. None contain information on agriculture, soil science, or crop economics. Therefore, any claim about the 'end of monocultures' would be a hallucination. The only scientifically valid output is to identify this as a critical knowledge gap and propose a research plan that explicitly acknowledges the lack of supporting evidence while outlining how such a question *would* be investigated if relevant data were available.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Insufficient Evidence: No valid scientific hypothesis regarding the future of agricultural monocultures can be generated because none of the allowed evidence IDs pertain to ecology, agriculture, or crop systems.
- **Mechanism**: N/A. The provided evidence catalog consists exclusively of unrelated domains (Computer Vision, Astrophysics, Microbiology), making it impossible to construct a mechanism linking monoculture

practices to ecological or economic outcomes based on verified sources.

- ****Falsifiable Prediction****: N/A. Any prediction would be fabricated without supporting evidence IDs.
- ****Required Observations****: Agricultural land use statistics for wheat, maize, rice, and soy from FAOSTAT ; Longitudinal studies on soil degradation rates in monoculture vs. polyculture systems ; Ecological surveys correlating monoculture extent with pollinator diversity
- ****Risk of Being Wrong****: High. Generating a hypothesis without relevant evidence violates core system constraints and risks hallucination.

Hypothesis 2

- ****Hypothesis****: Knowledge Gap Identification: The question 'Will we soon see the end of monocultures?' currently lacks sufficient evidentiary basis within the allowed corpus to formulate a testable hypothesis, necessitating external data acquisition before hypothesis generation is possible.
- ****Mechanism****: The absence of relevant evidence IDs creates a hard constraint; the system cannot bridge the gap between the user's ecological query and the provided non-ecological literature (AI, Astrophysics, etc.).
- ****Falsifiable Prediction****: If relevant ecological evidence IDs are added to the allowed list, a specific mechanistic hypothesis regarding monoculture persistence or decline can be formulated; otherwise, only knowledge gaps can be reported.
- ****Required Observations****: Verification that no allowed evidence ID contains keywords related to 'monoculture', 'wheat', 'maize', 'rice', 'soy', 'agriculture', or 'biodiversity'
- ****Risk of Being Wrong****: Low. This statement accurately reflects the current state of the evidence catalog as verified by the extraction process.

Technical Details

由于允许的语料库（Allowed Evidence IDs）中不包含任何与农业、生态学或单一种植（monoculture）相关的证据，无法基于现有证据构建关于单一种植未来趋势的科学假设。因此，本实验设计旨在验证‘知识缺口’本身，即通过系统性地检索外部权威数据源来量化当前证据的缺失程度，并建立基线数据管道，以便在未来引入相关证据时能够立即进行假设检验。技术路线包括：1. 定义关键生态与农业指标（如单一作物种植面积占比、土壤有机质变化率、传粉昆虫多样性指数）；2. 构建自动化数据获取脚本，从FAOSTAT和GBIF等公开数据库提取历史时间序列数据；3. 设计时空外推模型框架，用于在获得数据后预测极端事件（如单一种植系统的崩溃阈值）。

Datasets

Source

```
```json
```

```
[
```

```
{
```

```
"name": "FAOSTAT Global Agricultural Production Data",
```

```

"description": "Historical data on area harvested for wheat, maize, rice, and soy globally and by country.",
"url": "https://www.fao.org/faostat/en/#data/QC",
"license": "CC BY-NC-SA 3.0 IGO",
"evidence_ids": []
},
{
"name": "Global Biodiversity Information Facility (GBIF)",
"description": "Occurrence records for key pollinator species and soil health indicators.",
"url": "https://www.gbif.org/",
"license": "CC0 1.0",
"evidence_ids": []
}
]
'''

```

Target

```

'''json
{
"name": "Monoculture Viability Index Dataset",
"description": "A structured time-series dataset combining agricultural extent with ecological stressors, prepared
for extreme event prediction modeling.",
"format": "Parquet/CSV",
"features": [
"year",
"country_code",
"crop_type",
"area_ha",
"soil_degradation_index",
"pollinator_diversity_shannon",
"yield_volatility"
]
}
'''

```

}

'''

## Paper Abstract

Background: The question of whether major monocultures (wheat, maize, rice, soy) will end soon requires robust ecological and economic evidence. Methods: We systematically reviewed the allowed evidence catalog (4 EvidenceCards) for relevance to agriculture, soil science, or crop systems. Results: All provided evidence cards were found to be from unrelated domains (Computer Vision, Astrophysics, Microbiology). No established facts regarding monoculture trends could be extracted. Validation Plan: Future work requires the inclusion of FAOSTAT data and peer-reviewed ecological studies. Pending Results: No experimental results can be reported as no relevant data exists in the allowed set. Conclusion: The question cannot be answered with the current evidence constraints; this report serves as a formal identification of this critical knowledge gap.

## Methods

Systematic Evidence Relevance Screening. Each allowed EvidenceCard was analyzed for keywords related to 'monoculture', 'agriculture', 'soil', 'crop', or 'biodiversity'. All cards failed this screening. Consequently, no statistical modeling or hypothesis testing was performed on the provided evidence, as doing so would constitute scientific misconduct (fabrication).

## Experiments

### Baselines

'''json

[

"Naive Persistence Model: Assumes current monoculture trends continue linearly without ecological feedback.",

"Historical Average Baseline: Predicts future states based on the mean of the last 10 years of data.",

"Random Walk with Drift: Standard statistical baseline for non-stationary time series."

]

'''

### Metrics

'''json

[

"Top-K Recall (for identifying regions at highest risk of monoculture collapse)",

"Area Under Precision-Recall Curve (AUPRC) for rare extreme events",

"Brier Score for probabilistic forecasting of system state transitions",

"Root Mean Squared Error (RMSE) for continuous yield predictions"

]

...

## Ablation

Not applicable in the current phase as no model is trained. Future ablation will test the contribution of ecological variables (e.g., removing pollinator data) to prediction accuracy.

## Validation Protocol

Time-series cross-validation with expanding window. Training on data up to year T, validating on T+1 to T+5. Spatial hold-out validation will be used to test generalizability across different climatic zones.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。由于缺乏允许的证据ID支持，无法得出关于单一种植终结的任何事实性结论。

## References

- **\*\*EV-Q113-922f1ca507ca9ae55ea64686\*\*** · arxiv · arXiv:2312.03154
  - authors: Not available · year: Not available
  - url: <https://arxiv.org/pdf/2312.03154.pdf> · doi: Not available
  - reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=b2ae7e4699982ecd4b6fbd9a79026ce435ddc1251421f060b6e284dfe57aa99a
- **\*\*EV-Q113-9fbb7400332c8c3f75e4963c\*\*** · arxiv · arXiv:2503.17006
  - authors: Not available · year: Not available
  - url: <https://arxiv.org/pdf/2503.17006.pdf> · doi: Not available
  - reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=a88545e6c5dd8cd02c1dee0f0be139ae0a375512fe220fbd31f02b6b4348c5f2
- **\*\*EV-Q113-6eb7f4dea9363af7283d0969\*\*** · arxiv · arXiv:2402.05095
  - authors: Not available · year: Not available
  - url: <https://arxiv.org/pdf/2402.05095.pdf> · doi: Not available
  - reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=210901547e9967877fb8a41e4faca02c349314481b4f24317149513067456924

- **\*\*EV-Q113-f8776fe39ba76542f19b29e8\*\*** · arxiv · arXiv:2502.19555
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2502.19555.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=5a7acab3d33ca92b3544826c1043e9bdfae71e75e3d96a6940985409f21790a

## Reviewer Comments

- The candidate correctly identified that none of the allowed evidence IDs (EV-Q113-922f1ca507ca9ae55ea64686, EV-Q113-9fbb7400332c8c3f75e4963c, EV-Q113-6eb7f4dea9363af7283d0969, EV-Q113-f8776fe39ba76542f19b29e8) pertain to ecology, agriculture, or monocultures.
- The system appropriately refused to generate a scientific hypothesis about monoculture trends using unrelated papers on ControlNet, Stellar Tides, or Bacillus subtilis, adhering strictly to the no-hallucination constraint.
- The 'Knowledge Gap Identification' is the only valid output given the input constraints; treating this as a hypothesis regarding the state of the evidence corpus is methodologically sound for this edge case.
- Results are correctly marked as pending/not applicable, avoiding any fabrication of experimental outcomes.
- External data sources (FAOSTAT, GBIF) are correctly listed in datasets but not cited as established facts or evidence\_ids, maintaining strict separation between verified evidence and proposed future data.

## Revision History

## Reproducibility Checklist

- Verify that no allowed evidence ID was used to fabricate ecological facts.
- Confirm data extraction scripts point to public APIs (FAOSTAT, GBIF).
- Ensure all code is containerized using Docker for environment consistency.
- Document all data preprocessing steps, especially handling of missing values in historical records.
- Store random seeds for any stochastic components in future modeling phases.